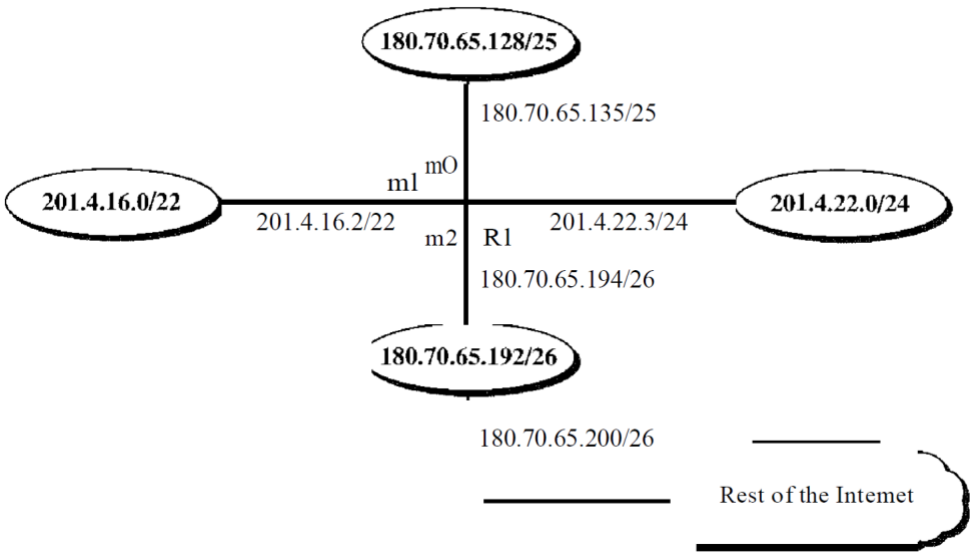
Tutorial 5 Topic6 Network Layer

Exercise 2



Make a routing table for router Rl, using the configuration in the Figure.

Show the forwarding process if a packet arrives at Rl with the destination address 180.70.65.140.

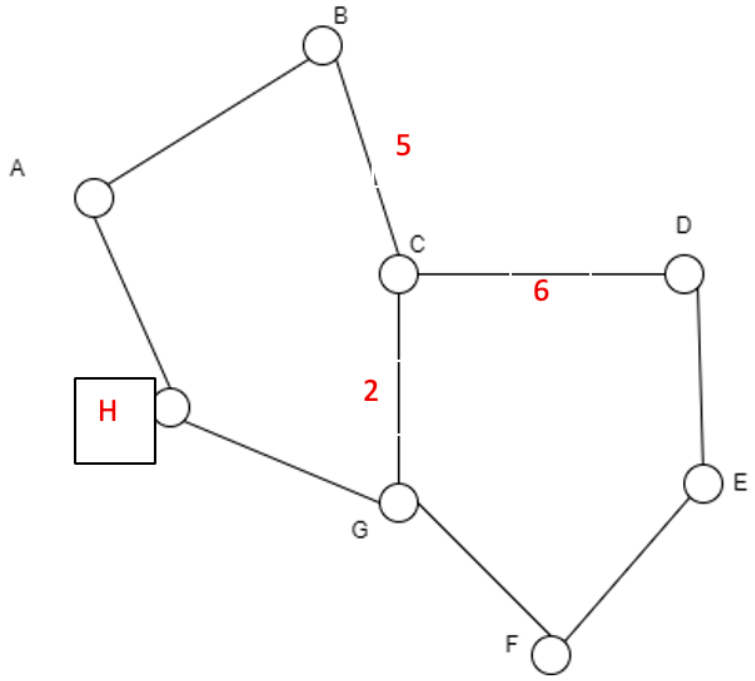
Exercise 3

Find the topology of the network if Table below is the routing table for a router R.

Mask	*Network Address*	*Next-Hop Address*	*Interface*
/27	202.14.17.224	-	rn1
/18	145.23.192.0	-	rnO
/0	0.0.0.0	130.56.12.4	m2

Exercise 4

Assume RIP is used as intra-AS protocol in the following AS. Node C receives the neighbors' distance vector. Show the new estimated vale of the distance vector in node C. Next, show the routing table in C.



C initial distance vector	Vectors Received from C Neighbors			New Estimated cost in C
	From B(+5)	From D(+6)	From G(+2)	
∞	2	8	4	$6\ (G)$
5	**0**	**10**	**6**	**5**
0	**5**	**6**	**2**	**0**
6	**10**	**0**	**4**	**6**
∞	9	1	3	$5(G)$
∞	8	2	2	$4(G)$
2	**6**	**4**	**0**	**2**
∞	3	7	3	$5(G)$

Show the routing table in C (don't show the interface)

Destination	Next Hop
A	**G**
B	**B**
D	**D**
E	**G**
F	**G**
G	**G**
H	**G**